

ASSESSMENT - CO1 AT3

NAME : PRENA M

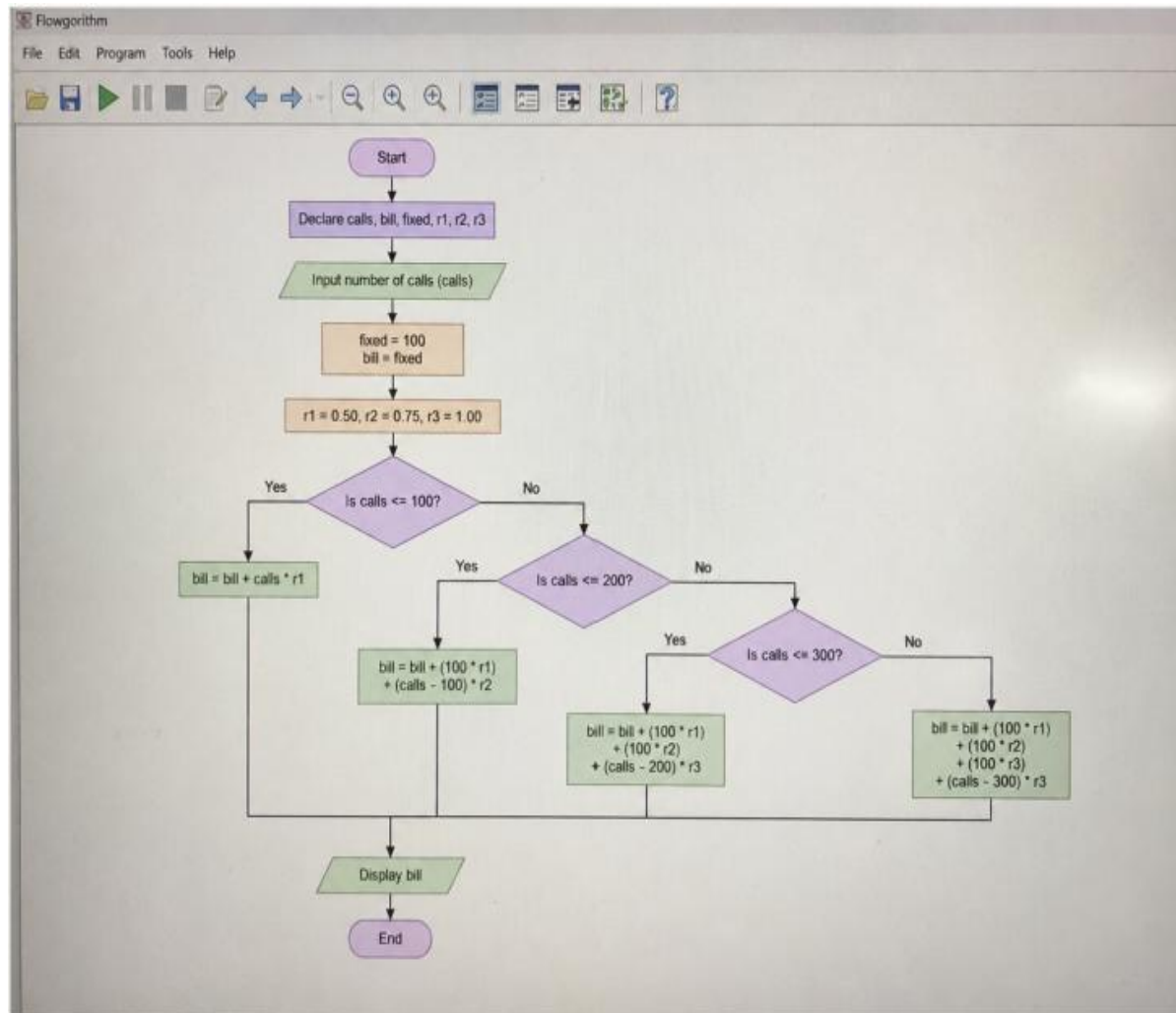
REG NO : 192524056

CODE : CSA 0924

COURSE : JAVA PROGRAMMING

1. Read the number of calls made and calculate a telephone bill with a fixed charge plus three usage slabs.

Flow chart:



Pseudo Code:

BEGIN

 DECLARE calls, bill

 DECLARE fixed = 100

 DECLARE rate1 = 0.50

 DECLARE rate2 = 0.75

 DECLARE rate3 =

1.00 INPUT calls bill

= fixed IF calls <= 100

THEN

 bill = bill + (calls * rate1)

ELSE IF calls <= 200 THEN

 bill = bill + (100 * rate1)

bill = bill + ((calls - 100) * rate2)

ELSE bill = bill + (100 * rate1)

bill = bill + (100 * rate2) bill =

bill + ((calls - 200) * rate3)

END IF

 OUTPUT "Telephone Bill = ", bill

END

Java program:

```
import java.util.Scanner; public class
TelephoneBill {    public static void
main(String[] args) {        Scanner sc =
new Scanner(System.in);
        int calls;        double
bill;        double fixed =
100;        double rate1 =
0.50;        double rate2 =
0.75;        double rate3 =
1.00;

        System.out.print("Enter the number of calls: ");
calls = sc.nextInt();        bill = fixed;        if (calls
<= 100) {                bill = bill + (calls * rate1);        }
else if (calls <= 200) {                bill = bill + (100 *
rate1)
                + ((calls - 100) * rate2);
        } else {                bill = bill +
(100 * rate1)                +
(100 * rate2)
                + ((calls - 200) * rate3);
        }

        System.out.println("Telephone Bill = " + bill);
sc.close();
    }
}
```

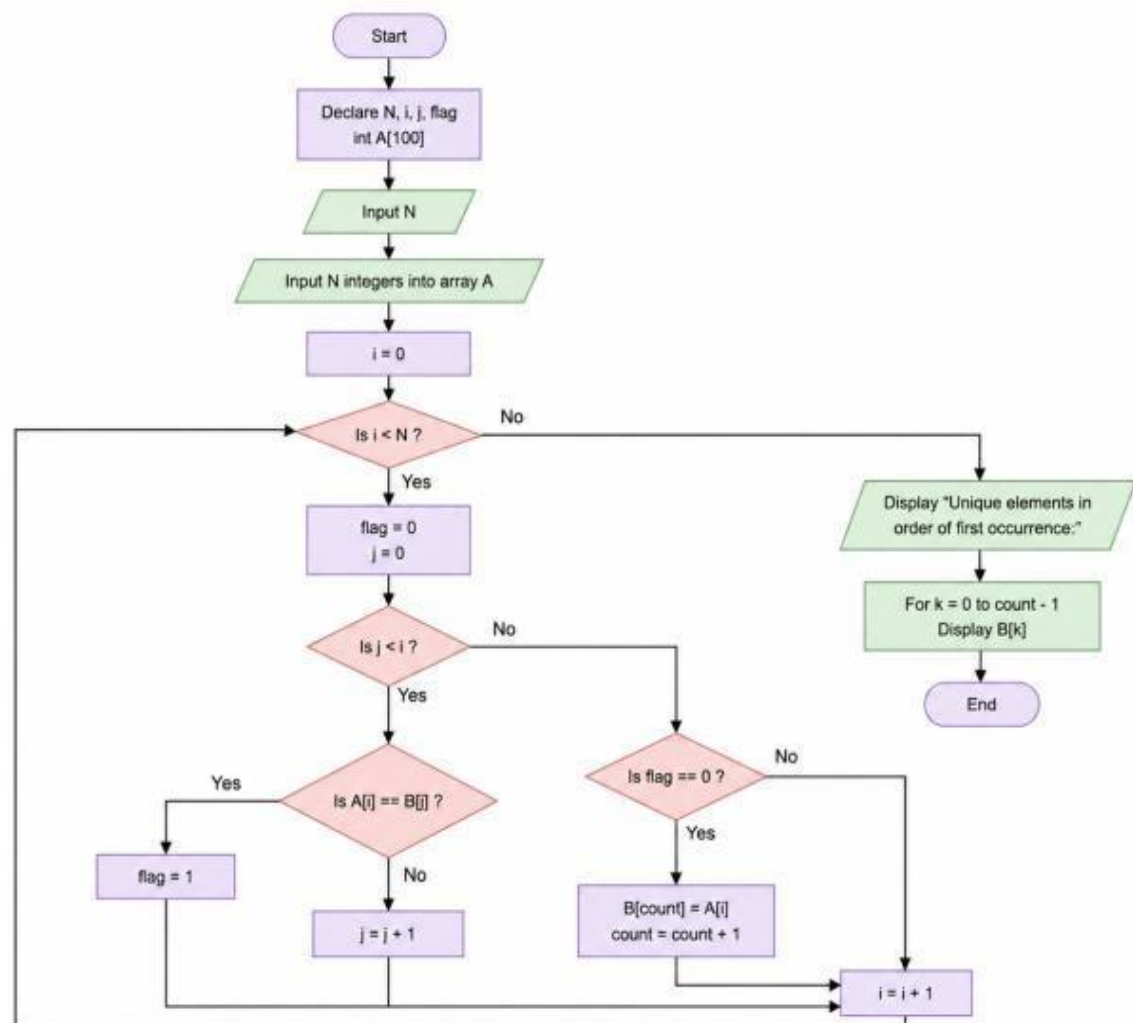
Sample Input:

Enter the number of calls: 250

Sample Output

Telephone Bill = 275.0

2. Read N integers and remove duplicate values while displaying only the first occurrence of each value.



Pseudo Code:

BEGIN

 Declare N, i, j

 Declare Array A[100]

 Declare flag

 INPUT N

 FOR i $\leftarrow$ 0 TO N-1

 INPUT A[i]

 END FOR

 OUTPUT "Unique Elements:" FOR

 i $\leftarrow$ 0 TO N-1

 flag $\leftarrow$ 0

 FOR j $\leftarrow$ 0 TO i-1

 IF A[i] = A[j] THEN

 flag $\leftarrow$ 1 BREAK

 END IF

 END FOR

 IF flag = 0 THEN

 OUTPUT A[i]

 END IF

 END FOR

END

Java program:

```
import java.util.Scanner; public class
RemoveDuplicates {    public static void
main(String[] args) {        Scanner sc = new
Scanner(System.in);

        int n;

        System.out.print("Enter the number of elements: ");

n = sc.nextInt();        int[] a = new int[n];

        System.out.println("Enter the elements:");

for (int i = 0; i < n; i++) {            a[i] = sc.nextInt();

        }

        System.out.println("Unique Elements:");

for (int i = 0; i < n; i++) {            boolean

flag = false;            for (int j = 0; j < i; j++) {

if (a[i] == a[j]) {                flag = true;

break;

            }

        }

if (!flag)

{

            System.out.print(a[i] + " ");

        }

    }

sc.close();

    }

}
```

Sample Input:

Enter the number of elements: 7

Enter the elements:

10 20 10 30 20 40 30

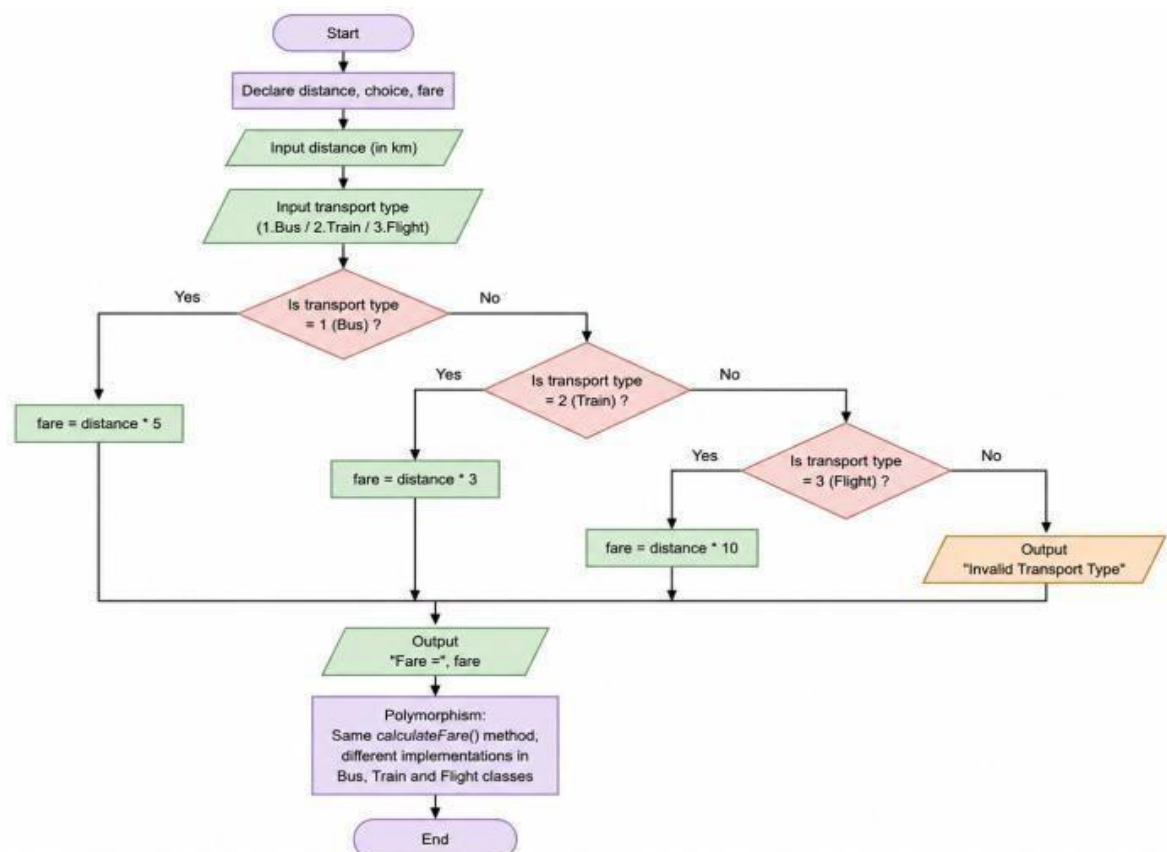
Sample Output:

Unique Elements:

10 20 30 40

3. Design a transport-fare flowchart for Bus, Train, and Flight. Use a common input process and different fare calculations, then relate it to polymorphism.

Flow chart:



Pseudo Code:

```
BEGIN

    DECLARE distance, fare

    DECLARE choice

    OUTPUT "Transport Fare Calculation"

    OUTPUT "1. Bus"

    OUTPUT "2. Train"

    OUTPUT "3. Flight"

    INPUT distance

    INPUT choice    IF
choice  =  1  THEN
fare ← distance × 5

    OUTPUT "Transport : Bus"
ELSE IF choice = 2 THEN    fare
← distance × 3

    OUTPUT "Transport : Train"
ELSE IF choice = 3 THEN
fare ← distance × 10

    OUTPUT "Transport : Flight"

ELSE

    OUTPUT "Invalid Transport Type"

    STOP

END IF

OUTPUT "Distance = ", distance

OUTPUT "Fare = ", fare

OUTPUT "Polymorphism:"

OUTPUT "Same method (calculateFare)"

OUTPUT "Different fare calculation"

OUTPUT "for Bus, Train and Flight."

END
```

Java code:

```
import java.util.Scanner; class
TransportFare {    public static void
main(String[] args) {    Scanner sc = new
Scanner(System.in);    int distance,
choice, fare = 0;

    System.out.print("Enter Distance: ");    distance = sc.nextInt();

    System.out.println("1.Bus 2.Train 3.Flight");

    System.out.print("Enter Choice: ");    choice =
sc.nextInt();    switch(choice) {    case 1:
fare = distance * 5;

        System.out.println("Transport: Bus");

        break;

    case 2:

        fare = distance * 3;

        System.out.println("Transport: Train");

        break;

    case 3:

        fare = distance * 10;

        System.out.println("Transport: Flight");

        break;

    default:

        System.out.println("Invalid Choice");

        return;

    }

    System.out.println("Fare = " + fare);

}
}
```

Input:

Enter Distance: 100

1. Bus
2. Train
3. Flight

Output:

Transport: Train

Fare = 300